

Homework 1

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Problem 1

(a) Final answer: 24

(b) $(1/2) w = [-2, 1, 3]^T$

$$v + (1/2) w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$\|v + (1/2) w\| = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3} \approx 5.20$$

Final answer: $3\sqrt{3}$

(c) $\cos \theta = (u \cdot w) / (\|u\| \|w\|)$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$\|u\| = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$\|w\| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$\cos \theta = 8 / (\sqrt{14} \cdot 2\sqrt{14}) = 8 / 28 = 2/7$$

$$\theta = \arccos(2/7) \approx 73^\circ$$

Final answer: $\theta \approx 73^\circ$

Problem 2

(a) AB : undefined, because A has 3 columns and B has only 2 rows ($3 \neq 2$).

BA is 2×3 since B is 2×3 and A is 3×3 .

row 1: $[1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$

row 2: $[3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$

$BA = [2 \ -9 \ -2]$

$[7 \ 4 \ -1]$

(b) $Av = [2(1) + (-1)(-2) + 0(3), 1(1) + 3(-2) + (-2)(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$

vA : undefined, v is 3×1 and A is 3×3 , so 1 column cannot meet 3 rows.

Final answer: $Av = [4, -11, -5]^T$, vA undefined

(c) Aw : undefined, A is 3×3 and w is 1×3 , so 3 columns cannot meet 1 row.

$wA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)] = [4, -6, -1]$

(d) vw is 3×1 times 1×3 , so it is 3×3 :

$vw = [2 \ 0 \ -1]$

$[-4 \ 0 \ 2]$

$[6 \ 0 \ -3]$

wv is 1×3 times 3×1 , so it is 1×1 : $2(1) + 0(-2) + (-1)(3) = -1$. $wv = [-1]$

(e) $A^2 = A$ times A :

row 1: $[2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$

$$\text{row 2: } [1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$$

$$\text{row 3: } [0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 16, -7]$$

$$A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$$

(f) B^2 undefined

Problem 3

$$(a) \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 2 & 3 & 1 & | & 2 \\ -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 2 & 3 & 1 & | & 2 \\ -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 2 & 3 & 1 & | & 2 \\ -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_2 \rightarrow -R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 10 & | & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 10 & | & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 10 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3: \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 1 & | & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 1 & | & 1 \end{bmatrix}$$

$$[0 \ 0 \ 1 \mid 1]$$

$x_3 = 1, x_2 = -4 + 3(1) = -1, x_1 = -1 - 2(-1) + 1 = 2$. One solution only.

Final answer: $x = [2, -1, 1]^T$

(b) $[1 \ 2 \ -1 \mid 4]$

$$[2 \ 5 \ 1 \mid 6]$$

$$[3 \ 7 \ 0 \mid 10]$$

$$R_2 \rightarrow R_2 - 2R_1: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[3 \ 7 \ 0 \mid 10]$$

$$R_3 \rightarrow R_3 - 3R_1: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

$$R_1 \rightarrow R_1 - 2R_2: [1 \ 0 \ -7 \mid 8]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

There is no pivot in column 3, so z is free.

Final answer: z is free, so there are infinitely many solutions

Problem 4

(a) Let x_1 = mass of Alloy 1 used, in kg; x_2 = mass of Alloy 2, in kg; x_3 = mass of

Alloy 3, in kg. The 100 kg of product holds 36 kg copper, 44 kg zinc, 20 kg nickel,

and none of the metal is lost, so in = out for each metal:

copper: $0.5x_1 + 0.2x_2 + 0.4x_3 = 36$

zinc: $0.3x_1 + 0.7x_2 + 0.2x_3 = 44$

nickel: $0.2x_1 + 0.1x_2 + 0.4x_3 = 20$

(b) Times 10 on every row to kill the decimals:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

(c) $R_1 \rightarrow R_1 - 2R_3$: $[1 \ 0 \ -4 \mid -40]$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$R_2 \rightarrow R_2 - 3R_1$: $[1 \ 0 \ -4 \mid -40]$

$$[0 \ 7 \ 14 \mid 560]$$

$$[2 \ 1 \ 4 \mid 200]$$

$R_3 \rightarrow R_3 - 2R_1$: $[1 \ 0 \ -4 \mid -40]$

$$[0 \ 7 \ 14 \mid 560]$$

$$[0 \ 1 \ 12 \mid 280]$$

$R_2 \rightarrow (1/7)R_2$: $[1 \ 0 \ -4 \mid -40]$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 10 \mid 200]$$

$$R_3 \rightarrow (1/10)R_3: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 1 \mid 20]$$

$$x_3 = 20, x_2 = 80 - 2(20) = 40, x_1 = -40 + 4(20) = 40$$

$$\text{Final answer: } x_1 = 40, x_2 = 40, x_3 = 20$$

Problem 5

$$(a) R_2 \rightarrow R_2 - 2R_1: [1 \ 2 \ 1]$$

$$[0 \ 0 \ 1]$$

$$[3 \ 6 \ 4]$$

$$R_3 \rightarrow R_3 - 3R_1: [1 \ 2 \ 1]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 1]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 2 \ 1]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 0]$$

$$R_1 \rightarrow R_1 - R_2: [1 \ 2 \ 0]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 0]$$

$$\text{Final answer: } \text{RREF}(A) = [1 \ 2 \ 0; 0 \ 0 \ 1; 0 \ 0 \ 0]$$

(b) There are pivots in column 1 and column 3 and nothing in row 3, so two pivots.

Final answer: $\text{rank}(A) = 2$

(c) Running the same row operations on $[A \mid b]$ turns the last row into

$$[0 \ 0 \ 0 \mid b_3 - b_1 - b_2].$$

0 solutions is possible: pick $b = [0, 0, 1]^T$, then that last row says $0 = 1$, which is

impossible, so the system is inconsistent.

Exactly 1 solution is impossible: column 2 has no pivot, so x_2 is always free. Any time

the system is consistent that free variable gives a whole family of answers. Put another

way $\text{rank}(A) = 2$ but there are 3 unknowns, and $2 < 3$, so a unique solution can't happen.

Infinitely many is possible: take $b = [1, 3, 4]^T$ (or $b = 0$). Then $b_3 = b_1 + b_2$, the last

row is $0 = 0$, the system is consistent, and free x_2 gives infinitely many solutions.

Problem 6

(a) These are dependent because there are three vectors and v_3 has a zero entry, so they

can't fill $\mathbb{R}^3$. The span is a plane.

(b) None of these is a multiple of another so they are independent; span is $\mathbb{R}^3$.
